

Installing Django

To install Django, use the following commands:

```
sudo pip3 install django
sudo pip3 install django --upgrade
```

Installing and configuring MongoDB

Install MongoDB by using the following code:

```
sudo apt-get install mongodb
sudo service mongodb start
```

MongoDB security configuration is:

```
mongod --port 27017 --dbpath /var/lib/
mongodb
```

The command given above will start the *mongodb* service and then you can open the Mongo console by giving the following command:

```
mongo --port 27017
```

Create an admin user as follows:

```
use admin
db.createUser(
  {
    user: "admin",
    pwd: "<pwd here>",
    roles: [ { role:
"userAdminAnyDatabase", db: "admin" },
"readWriteAnyDatabase" ]
  }
);
```

Granting root privileges to the user admin

This is essential for managing other users. To do so, use the following code:

```
db.grantRolesToUser('admin', [{ role:
'root', db: 'admin' }])
```

Shut down db to open in auth mode as follows:

```
db.adminCommand( { shutdown: 1 } )
```

Opening MongoDB in auth-enabled mode

To do so, use the code given below:

```
sudo mongod --auth --port 27017
--dbpath /var/lib/mongodb --bind_ip_all
```

--auth will enable Mongo in auth mode and --bind_ip_all makes Mongo host accessible from outside.

```
sudo mongo --port 27017 -u "admin"
--authenticationDatabase "admin"
# it will prompt for admin password
use admin
```

Creating application user

You are going to use this user from your Web app:

```
db.createUser(
  {
    user: "appuser",
    pwd: "<pwd>",
    roles: [ { role: "readWrite", db:
"<db name>" } ]
  }
);
```

```
use <app db name>
# creating application user - you are
going to use this user from your web
app,
db.createUser(
  {
    user: "appuser",
    pwd: "<pwd>",
    roles: [ { role: "readWrite", db:
"<db name>" } ]
  }
);
```

The command given below restores data from your local DB to GCP.

```
mongorestore --username admin
--password <admin_pwd_here>
--authenticationDatabase admin -d <DB_
name_to_import_data> <folder_name_
where_data_exported>
```

Creating Ubuntu user

```
adduser <user_name here>
```

The command given above will prompt you to enter required details like name, password, etc.

The command given below is to enable sudo access to the newly created user:

```
sudo usermod -aG sudo <user_name
here>
```

The command given below switches the user to a newly created user session:

```
sudo su <user_name here>
```

You can use this Ubuntu user for storing your application files, etc, and also for managing application deployments.

Running your Django application

Copy Django code into the server by *scp* or by using Filezilla, etc. From your project root folder, run the command given below to start the application:

```
/usr/bin/python3 manage.py runserver
0.0.0.0:8000
```

IP 0.0.0.0 is for accessing application with the right IP of the server. You may specify * at *ALLOWED_HOSTS* in Django settings.

You can now access your Django application by directly typing the URL in your browser, i.e., *http://VM_IP:8000*.

The next article in this series will cover Django application production deployment with Supervisor, SSL and Nginx. 

 By: Somanath Balakrishnan

The author has 16+ years of experience in the IT industry and is currently working as a technology consultant at Digital Core Technologies (<http://dicortech.com>), Kochi, Kerala.